DP Biology – D4.2 (HL extension)

Succession

- What allows ecosystems to remain stable indefinitely?
- Which human changes put ecosystem stability at risk?

HL statements D4.2.11–14. Textbook pp. 814–819.

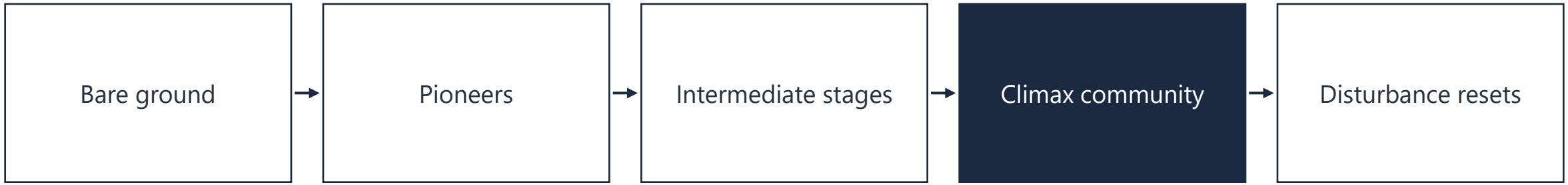


Image: lichen on bare rock. Daniel Schwen, CC BY-SA 4.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	D4.2.11	Explain what ecological succession is and what drives it
2	D4.2.12	Describe the changes that take place during primary succession
3	D4.2.13–14	Explain cyclical succession, climax communities and arrested succession

The big picture: communities change over time



Each community changes conditions so that the **next** community can establish. Humans can halt or reset the process.

Big question

Why does an abandoned field turn into woodland, but grazed pasture does not?

Ecological succession

D4.2.11

Succession is a directional change in the species of a community over time.

Driven by **abiotic** changes (water, soil, temperature, fire) and **biotic** ones (competition, herbivores).

Type	Starts on
Primary	Bare rock, lava, sand: no soil
Secondary	Soil left after fire, flood or clearing

Try it

Classify: an abandoned field; a new volcanic island.

Key idea

Both types move from pioneers, through intermediate stages, towards a climax.

What drives each change?

D4.2.11

Abiotic changes

Soil forms and deepens; more water held; shade lowers light and temperature at ground level

Biotic changes

Taller plants outcompete smaller ones; new herbivores and predators arrive; food webs grow

Key idea

Organisms change their environment, and the changed environment favours different organisms.

Primary succession

D4.2.12

Bare rock: no soil → **Pioneers: lichens, mosses** → Thin soil: grasses, herbs → Deeper soil: shrubs, trees → Climax: e.g. forest



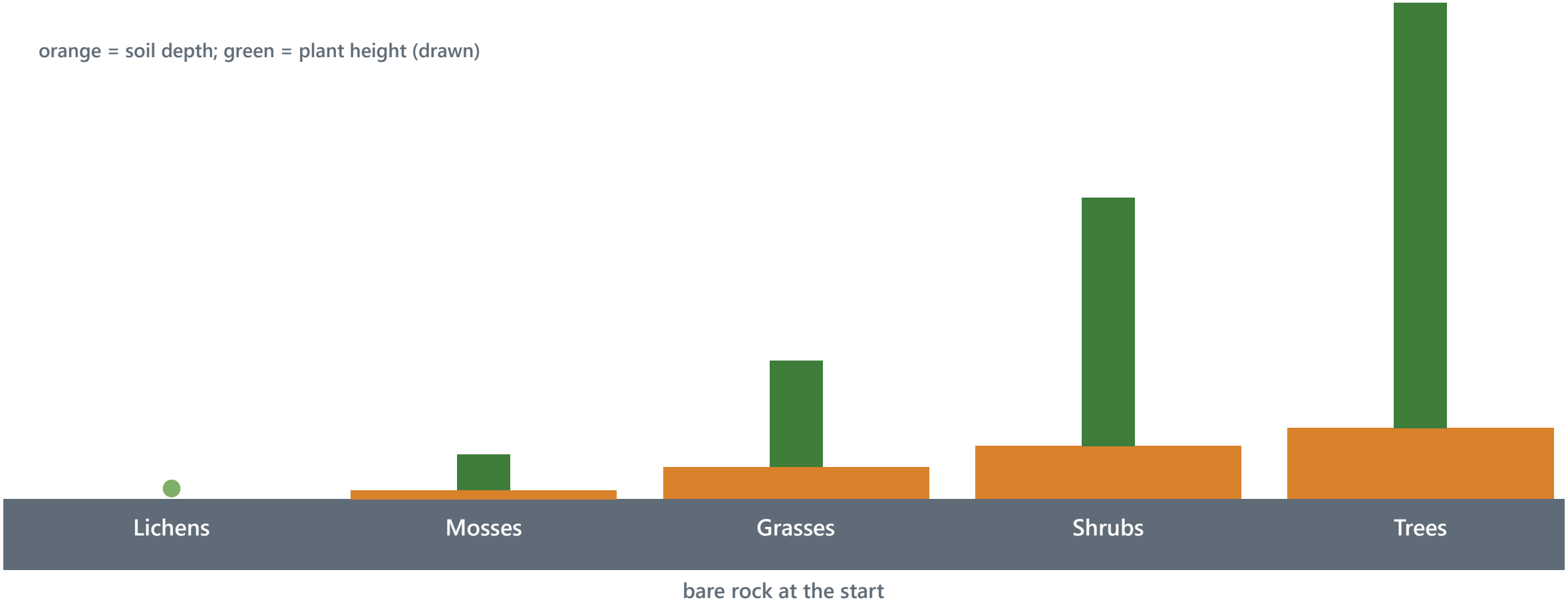
A pioneer on bare rock. Daniel Schwen, CC BY-SA 4.0.

Each stage helps the next

Pioneers trap dust and add dead matter, forming soil. Soil holds water and minerals so larger plants root; their shade changes conditions again and they outcompete earlier species.

Soil builds, plants grow

D4.2.12



Check-in: class 1

D4.2.11–12

1. Define succession.
2. Distinguish primary and secondary succession.
3. Give an abiotic and a biotic driver.
4. Explain the role of pioneers.
5. List the stages of primary succession.

How to use this

Answer without your notes, then check against slides 3–7.

What increases during primary succession

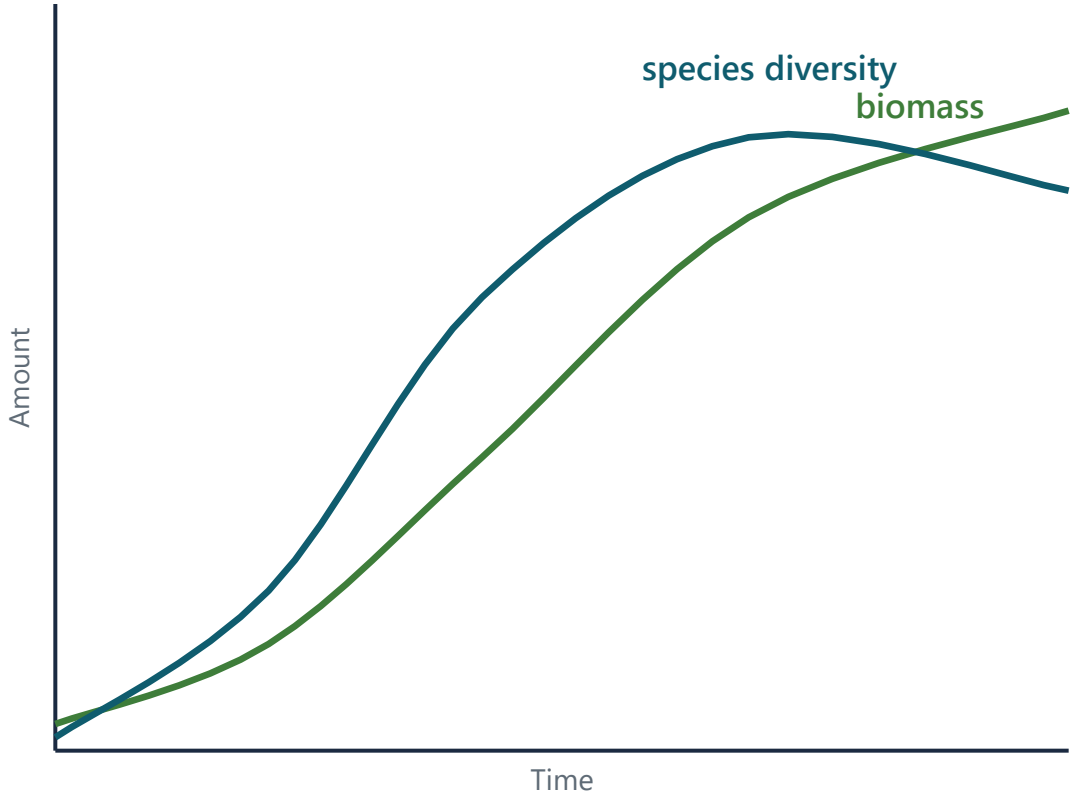
D4.2.12

Feature	Early	Late
Plant size	Small	Large trees
Primary production	Low	High
Species diversity	Low	High
Food web complexity	Short chains	Many links
Nutrient cycling	Little; easily lost	Much; held and recycled

Try it

Sketch soil depth and species number against time.

Trends over time



Sketch of the general pattern.

Why diversity may dip

In a late climax, a few dominant species can shade out others, so diversity may fall slightly.

Secondary succession

D4.2.11–12

Where soil remains (fire, flood, abandoned field), succession is **faster**: seeds, roots and soil organisms remain.

Stage	Abandoned field	After a forest fire
Early	Annual weeds, grasses	Herbs; seeds released by heat
Middle	Shrubs, fast-growing trees	Shrubs, sprouting roots
Late	Shade-tolerant trees	Mature forest after decades

Working with data: sand dunes

D4.2.12

Youngest dunes are nearest the sea: walking inland is like moving forward in time. Invented values for practice.

Inland / m	Species	Organic matter / %	Soil depth / cm
10	3	0.5	1
100	12	2	5
300	25	6	15
600	30	12	30

1. Describe two trends.
2. Explain how early plants help later ones.
3. Why are dunes useful for studying succession?

Practice: succession trends

D4.2.12

Explain

Why food webs grow more complex.

Explain

Why nutrients are lost early on.

Compare

Primary and secondary succession speed.

Use

pioneer, biomass, nutrient cycling, climax

Watching succession

D4.2.11–12

After the 1980 Mount St Helens eruption, scientists tracked primary succession on fresh ash: lupins fixed nitrogen and sped the process.

Discuss

How could you study a process that takes centuries? (Chronosequences such as dunes or glacier forelands; long-term plots.)

Check-in: class 2

D4.2.12

1. Give five trends in primary succession.
2. Explain why secondary succession is faster.
3. Explain why soil depth increases.
4. Explain a chronosequence.
5. Give an example of a nitrogen-fixing pioneer.

How to use this

Answer without your notes, then check against slides 9–14.

Cyclical succession

D4.2.13



Barnacles on a rocky shore. Stemonitis, free use.

Some communities replace each other in a **repeating cycle** with no fixed climax.

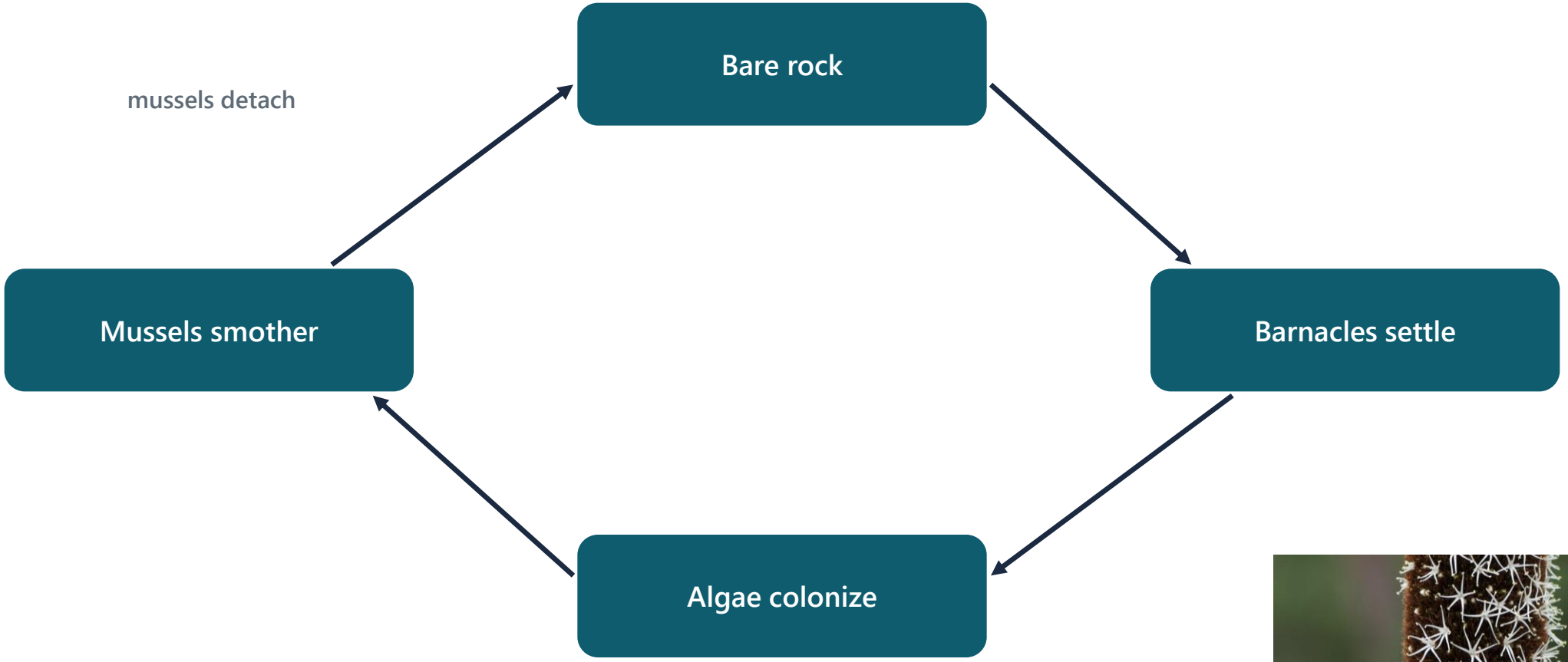
New Zealand rocky shores: barnacles, then algae, then mussels; mussels fall off and the cycle restarts.

Also

Regular fires reset Australian bush; grass trees flower soon after burning.

A cycle on the shore

D4.2.13





Grass tree. Eric in SF, CC BY-SA 3.0.

Climax communities

D4.2.14

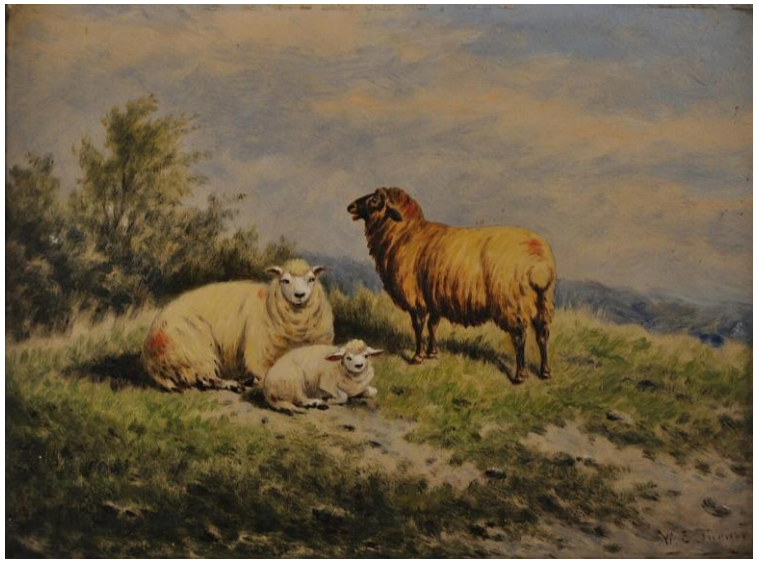
A **climax community** is stable and in balance with its climate and soil, until disturbed. Its type depends on climate: forest, grassland or tundra.

Common mistake

Succession does not always end in forest: in dry or cold climates the climax may be grassland or tundra (link B4.1 biomes).

Arrested succession

D4.2.14



Sheep at pasture, painting. William E. Turner, public domain.

Grazing

Livestock eat tree and shrub seedlings, so grassland stays where forest would grow.

Draining wetlands

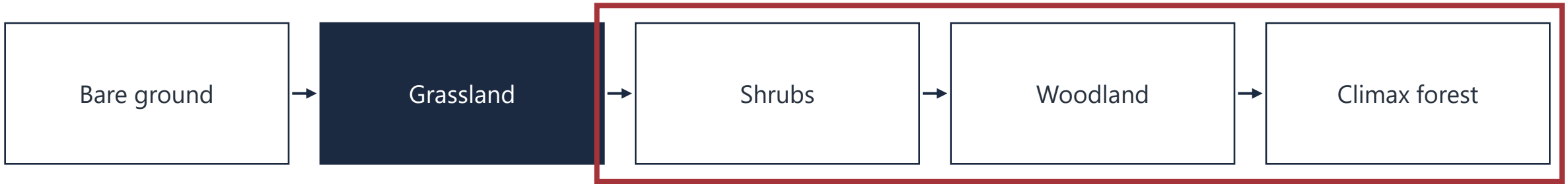
Drying soil removes waterlogged habitat; the natural community cannot develop.

Big question

Stop the pressure and succession restarts: why rewilding removes grazers or blocks drains.

Stopped at grassland

D4.2.14



Examples

Alpine pastures kept open by cows; heathland kept by burning; drained peat bogs used for farming.

Working with data: removing grazers

D4.2.14

Tree seedlings per 100 m^2 inside and outside a fence that keeps deer out. Invented values; realistic pattern.

Years after fencing	Inside fence	Outside fence
0	2	2
3	15	3
6	40	2
10	65	3

1. Compare the trends.
2. Explain the difference.
3. Predict the inside plot in 50 years.

Check-in: class 3

D4.2.13–14

1. Define cyclical succession.
2. Give an example.
3. Define climax community.
4. Explain how grazing arrests succession.
5. Explain how draining wetlands arrests succession.

How to use this

Answer without your notes, then check against slides 16–21.

Common mistakes to avoid

Not this	But this
Succession always ends in forest.	The climax depends on climate.
Pioneers stay forever.	They are replaced.
Secondary succession starts with no soil.	Soil remains.

Summary: succession

D4.2.11–14

	Key idea
D4.2.11	Succession: directional change; abiotic and biotic drivers
D4.2.12	Primary succession and its trends
D4.2.13	Cyclical succession
D4.2.14	Climax and arrested succession

Check that you can...

1. **Distinguish** primary and secondary succession
2. **Outline** changes during primary succession
3. **Explain** cyclical succession
4. **Explain** arrested succession

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Distinguish between primary and secondary succession.
2. Explain the role of pioneer species.
3. Explain how grazing livestock arrests succession.

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Sheep grazing.jpg – William Eddowes Turner – Public domain

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